

Wireshark Analysis, TCP Handshakes, Real Traffic

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 1 hour.

TCP Three-Way Handshake

Remember: SYN, SYN/ACK, ACK

What problem does the TCP handshake solve?

It solves the problem of ensuring both sides of a network connection are ready, synchronized, and agree to communicate before any data is sent. Without it, there's no guarantee that the sender and receiver are both online, listening, and prepared to exchange data.

What would happen if TCP did NOT confirm connections?

Data could be sent to devices that aren't ready. Lost packets would go undetected. Connections would become unreliable and security problems could increase.

Why is sequence numbering important in TCP?

Well, in TCP every byte of data sent across the network is given a sequence number. When data travels across the internet: packets can arrive out of order, some packets may get lost, and some may even be duplicated. Therefore, it helps TCP put packets back in the correct order, detect missing packets, identify duplicate packets, and ensures all data arrives correctly. Without it, files could become corrupted, webpages could load incorrectly, videos and downloads could break, and communication would become unreliable.

More Wireshark Analysis

Open Wireshark. Start the capture. Open Edge, Youtube, Discord, Tabs, etc. Then filter: **dns**.

Look for: query names, destination domains, and source / destination IPs.

What is a DNS query vs DNS response?

A DNS query is the request asking: "What is the IP address for this domain name?" My computer sends the query to a DNS server. Example: I type youtube.com. The computer sends a DNS query asking: "What IP address belongs to youtube.com?"

A DNS response is the answer returned by the DNS server.

Now I am filtering for **tcp**. I am looking at source IP, destination IP, source port, destination port, sequence numbers, and flags.

What are TCP flags?

Small control bits inside a TCP packet that tell the other computer what kind of action is happening during the connection. (SYN, ACK, FIN, RST)

SYN: used to start a TCP connection

ACK: confirm that data was received successfully

FIN: gracefully close a connection

RST: immediately terminates a connection

TLS + Encrypted Traffic

Filter by **tls**.

What information is sent in a TLS Client Hello?

The TLS Client Hello is the very first message sent by a client when starting a secure HTTPS/TLS connection with a server.

Basically says: "Hi, I want to communicate securely. Here are the security options I support."

Info Sent:

1. TLS Version: TLS 1.2? TLS 1.3?
2. Random Number: The client sends a random large value. This random number is later used to help generate: encryption keys, session keys, and secure communication secrets.
3. Cipher Suites: The client lists the encryption algorithm it supports. These are called cipher suites. (AES, ChaCha20, RSA, ECDHE)